PATIENT CARE THEORY 2

UNIT 7, PART 4: Other Pediatric Illness

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Learning Objectives

- Identify common pediatric illness and communicable disease

Dehydration

- Dehydration results in fluid loss at a slower pace
  - In children dehydration can lead to serious volume depletion quickly compared with adults
- Typical causes
  - Vomiting, diarrhea, diabetic ketoacidosis
- Electrolyte imbalance is common
- Potential for arrhythmias
- Altered LOC from volume depletion, electrolyte and/or acid-base imbalance
- Signs of dehydration?

Dehydration

Management

- Supportive Care
- ABCDE
- Minimal scene time, maintain warmth
- Rapid transport

Enroute

- For hypotension…
  - IV access & fluid resuscitation 20 cc/kg of NS (slow)
- Caveat: aggressive/rapid fluid resuscitation can result in cerebral edema in patients with diabetic ketoacidosis

Infectious Diseases

- Common Childhood Infections
  - RSV
  - Thrush
  - Meningitis
  - Norovirus (most common in childcare centers; H/a, sore throat, N/V/D)
  - Chicken Pox
  - Conjunctivitis (Pink eye)
  - Fifth disease
  - Coxsackie Disease (hand, foot, mouth)
  - Impetigo
  - Roseola
  - Scarlett Fever (Strep infections)
  - And many more…….

Concerns in Pregnancy

Many pregnant women will have small children at home

- Chicken pox
- Shingles (reactivation of Chicken pox)
- CMV
- Fifth Disease (slapped cheek)
- Listeriosis
- Toxoplasmosis

Meningitis

- Inflammation or infection of the meninges
- Bacterial or Viral
  - Which is more dangerous?
- Symptoms vary by age
  - Fever
  - Altered LOA
  - Bulging fontanelle
  - +/- nuchal rigidity and photophobia
  - Petechial or purpuric rash (***non-blanching***)
  - Signs of shock

Seizures

- Pediatric seizures are common (~5% will have a seizure by 16 yrs of age)
- Seizure or seizure mimic?
- Elements that are highly suggestive of true seizure
  - Lateralized tongue biting (high specificity)
  - Flickering eyelids
  - Dilated pupils with blank stare
  - Lip smacking
  - Increased HR and BP ***during*** event
  - Post-ictal phase
- Neonates are more likely to present with subtle seizures that lack the "tonic-clonic" activity

Seizures

- Types of Seizures
  - Generalized
    - Involve loss of awareness and both hemispheres of the brain
    - Tonic-clonic or Grand mal seizure; Absence seizure
  - Focal
    - May involve loss of awareness and only involve one part of the brain
    - Focal aware – may include jerking movements but still conscious
    - Focal impaired awareness – focal motor jerking with loss of awareness

Seizures

- Breath-holding
  - Common in 6-18 month old
  - Usually a clear trigger (emotional distress, pain)
  - LOC and breath-holding can produce seizure like activity (short duration from decrease cerebral blood flow) however, recovery is rapid (no post ictal phase)
  - Syncopal episode

Febrile Seizure

	Simple Febrile Seizure	Complex Febrile Seizure
Age	6mos to 5yrs	Any
Frequency	Single seizure in 24hrs	Multiple seizures in 24hrs
Nature	Generalized	Focal or generalized
Duration	Lasting < 15 min	Can be prolonged
Recovery	Post- ictal with return to baseline and normal neurological exam	Post- ictal, may not fully return to normal if multiple seizures

Febrile Seizures

Simple febrile seizures tend to occur early in the illness within 24hrs of onset of fever –

if the seizure occurs >24hrs after the onset of fever, the suspicion for a bacterial cause of the fever

and a pathologic cause for the seizure should be heightened.

Febrile Seizures

Parents will often blame themselves for not treating the seizure appropriately or quickly enough to prevent the seizure. It is important to educate them that the height of the fever or rapidity of onset does not predict risk of suffering a seizure and that prophylactic antipyretics do not decrease the rate of seizure recurrence.

Assessment/Management - Seizures

- General Impression
- ABCDE
- Most patients will be post-ictal on your arrival
  - If not consider time (> 10 minutes assume status epilepticus)
- Supportive care
  - Do not try to force anything in the mouth – manual positioning/ NPA
  - O2
  - PPV
  - ACP

SIDS (Sudden Infant Death Syndrome)

- Unknown etiology
  - Unexplained death of an infant <1 year of age (peak @ 2-4 months)
  - Thought to be related to an underlying vulnerability or underdeveloped respiratory control center and sleep-wake regulation
  - May require initiation of resuscitation despite obvious signs of death
  - Provide emotional support
  - Obtain as much history as possible (time permitting) and visualize space where infant was sleeping/found
  - Ensure police response

SIDS

Risk Factors:

- Premature
- Low birth rate
- Males (slightly higher risk)
- Family history
- Recent respiratory infection
- Smoking during pregnancy or exposure to second-hand smoke

Toxicological Emergencies

- More than 50% of toxic exposures are seen in children < 6 yrs
  - Ingestion
  - Inhalation
  - Injection
  - Absorption
- Consider age of child
  - < 5 yrs likely accidental/ unintentional

Poisoning

- Calcium channel blocker or beta blocker ingestion can cause hypotension and weakened cardiac contractility or acute heart failure
- As little as 2 Glyburide tablets (5 mg ea.) can be fatal to a toddler

Management

Supportive Care

- ABCDE (include BGL early)
- Focused and thorough history (identify agents as quickly as possible)
- Minimal scene time – consider contacting Poison Control Center
- Toxidromes
- Decontamination and/or elimination and Antidotes
- Rapid transport
- IV access & fluids

ACP – “Antidotes”

- +/- high dose IV glucagon (BB OD), NaHCO-3, atropine, etc.

Child Abuse

- “Child abuse” includes physical, emotional and sexual abuse and/or neglect. (Ontario association of children’s aid societies)
- Physical:
  - Child is injured or harmed by his or her caregiver or when caregiver fails to do something to protect the child
    - Child wearing long sleeves/pants even in warm weather
      - Patterned bruising, burns, hand prints etc
    - Excessive crying
    - Avoidance of physical contact with others
    - Whispered speech
    - Wary of adults
    - clinging

Child Abuse

- Emotional:
  - Caregiver treats a child in an extremely negative way that damages self-esteem and the concept of self
    - Headaches, stomach aches, no medical cause
    - Abnormal fears/nightmares
    - Desperately affectionate behaviours
    - Speech disorders
    - Habit disorders (biting, rocking, head-banging)
    - Constant tantrums
    - Extreme behaviours – disobedient or overly compliant

Child Abuse

- Sexual Abuse:
  - Sexual exploitation of a child by a caregiver or someone else
    - Frequent sore throats or UTI
    - Constant sadness
    - Thumb sucking
    - Aggressiveness or withdrawal
    - Avoidance of undressing
    - bedwetting

Child Abuse

- Sexual Abuse
  - Older children (school-aged, teens)
  - Consider human trafficking when
    - Frequent EMS calls
    - Substance abuse
    - Incongruent behaviour

Child Abuse

- Neglect
  - A pattern where a child's caregiver fails to provide the basic needs (food, sleep, safety, shelter, supervision etc.)
    - Missing key items of clothing
    - Over or under-dressed for weather conditions
    - Persistent hunger
    - Body odour
    - Child assumes adult responsibilities
    - Always tired
    - Stealing food/money from others

Child Abuse

- The previous lists are by no means exhaustive
- Legal requirements for suspected abuse?
- Child in Need of Protection Standard (Training Bulletin 116- Paramedic Practice Documents)
  - section 72 of the CFSA legislates that anyone with reasonable grounds to suspect child abuse and/or neglect must immediately report the suspicion and the information on which it is based directly to a Children's Aid Society (CAS). This is known as the "duty to report".
  - Failure to report can result in fine up to $1000
  - This duty overrides the PHIPA

Management

- Begin as you would any other pediatric assessment
- Keep abuse in mind…..
- Hands-off approach; PAT
- ABCDE
- SpO2 – O2 PRN
- Treat any life threatening injuries
- Initiate transport
- Do not confront caregivers with your concerns
- ***Document*** any inconsistencies in stories/HPI

Reference Chart

Pediatric Vital Signs Reference Chart

Heart Rate (beats/min)			Respiratory Rate (breaths/min)	
Age	**Awake**	**Asleep**	**Age**	**Normal**
Neonate (<28 d)	100-205	90-160	Infant (<1 y)	30-53
Infant (1-12 mos)	100-190			
Toddler (1-2 y)	98-140	80-120	Toddler (1-2 y)	22-37
Preschool (3-5 y)	80-120	65-100	Preschool (3-5 y)	20-28
School-age (6-11 y)	75-118	58-90	School-age (6-11 y)	18-25
Adolescent (12-15 y)	60-100	50-90	Adolescent (12-15 y)	12-20

Reference: PALS Guidelines, 2015

Blood Pressure (mmHg)				
Age		**Systolic**	**Diastolic**	**Systolic Hypotension**
Birth (12 h)	<1 kg	39-59	16-36	<40-50
	3 kg	60-76	31-45	<50
Neonate (96 h)		67-84	35-53	<60
Infant (1-12 mos)		72-104	37-56	<70
Toddler (1-2 y)		86-106	42-63	<70 + (age in years × 2)
Preschool (3-5 y)		89-112	46-72	
School-age (6-9 y)		97-115	57-76	
Preadolescent (10-11 y)		102-120	61-80	<90
Adolescent (12-15 y)		110-131	64-83	

Reference: PALS Guidelines, 2015
For diagnosis of hypertension, refer to the 2017 AAP guidelines Table 4 & 5:
http://pediatrics.aappublications.org/content/early/2017/08/21/peds.2017-1904

Temperature (°C)	
Method	**Normal**
Rectal	36.6-38.0
Tympanic	35.8-38.0
Oral	35.5-37.5
Axillary	36.5-37.5

Ranges do not vary with age.
Screening: axillary, temporal, tympanic (↓ accuracy)
Definitive: rectal & oral (↑ reflection of core temp.)
Reference: CPS Position Statement on Temperature Measurement in Pediatrics (2015)

Oxygen Saturation (SpO_2)

SpO_2 is lower in the immediate newborn period. Beyond this period, a SpO_2 of **<90-92%** may suggest a **respiratory condition** or **cyanotic heart disease**.

Dr. Chris Novak & Dr. Peter Gill for www.pedscases.com
(Edited March 2020 by Richard He)

Summary – Key Points

- Children are not just little adults
  - Acknowledge and understand the anatomical and developmental differences
  - Get down to their level
- Use the PAT
  - Perform at the same time as you attempt to get information from parents and caregivers
  - Allow parents or caregivers to remain with the child whenever appropriate
- Use favorite toys to assist; perform assessments on parents or stuffed animals first
- Listen to parents – if a parent tells you that something just isn't right, believe them. They know their child better than you
- Know your normal vitals signs for each age group

Summary – Key points

- Know developmental markers by age:
- Infants (0-12 months) generally respond to the voice or face of their parents; like to be held by caregivers; crying can indicate pain, discomfort or hunger.
- Toddlers (1-3 years) are curious- more apt to have an ingestion emergency or FBAO. Toddlers fear separation from their parents, using stuffed animals and allowing them to sit on their parents' lap might help build trust.
- Preschoolers (3-5 years) can talk with simple words, but often can't understand what's happening and are scared by the sight of blood, so it's important to bandage even the simplest cuts and give constant reassurance.
- School-aged kids (6-12 years) can generally answer questions and follow instruction but have very vivid imaginations, especially about death, constant reminders that they'll be okay may be needed
- Adolescents (13-18 years) can provide accurate information but fear permanent scarring with trauma, feel modesty is very important to them, and can get caught up in the hysteria of a 911 call, so it's important to be well versed in a variety of calming measures.
- Pediatric assessment tips for EMTs, paramedics, nurses (ems1.com)

Pediatric GCS

Adults, children over 2 years	SCORE	Under 2 years
Eye Opening Response		
Spontaneous — opens with blinking at baseline	4 points	Eye opening spontaneously
Opens to verbal command, speech or shout	3 points	Eye opening to speech
Opens to pain	2 points	Eye opening to pain
None	1 point	No eye opening

Pediatric GCS

Verbal Response		
Oriented and converses	5 points	Infant coos or babbles (normal activity)
Confused, but able to answer questions	4 points	Infant is irritable and continually cries
Inappropriate responses, words are discernible	3 points	Infant cries to pain
Incomprehensible speech / sounds	2 points	Infant moans to pain
None	1 point	No verbal response

Pediatric GCS

Motor Response		
Obeys commands for movement	6 points	Infant moves spontaneously or purposefully
Purposeful movement to painful stimulus	5 points	Infant withdraws from touch
Withdraws from pain	4 points	Infant withdraws from pain
Abnormal (spastic) flexion, decorticate posture	3 points	Abnormal flexion to pain for an infant (decorticate response)
Extensor (rigid) response, decerebrate posture	2 points	Extension to pain (decerebrate response)
None	1 point	No motor response

Questions????